





Smart Donation Connect

AN AI-ENABLED PLATFORM FOR REDUCING WASTE AND CONNECTING DONORS WITH PEOPLE IN NEED..

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Abstract..

Smart Donation Connect is an AI-enabled platform that connects donors with verified people and organizations in need. It supports donations such as surplus food, clothes, electronic items, and money. The system uses AI/ML-based food-demand prediction to identify potential surplus food and a smart matching system to connect donors and receivers based on location, requirements, quantity, and urgency. The platform also provides notifications and donation tracking. The main aim is to reduce resource wastage and make the donation process more efficient, transparent, and accessible.



PROBLEM STATEMENT...

¤ Large amounts of food, clothing, electronics, and other useful resources are wasted while many people are in need.

¤ Hotels and restaurants often have surplus food, but it is difficult to identify nearby genuine receivers quickly.

¤ Donors may struggle to find verified people or organizations that actually need their support.

¤ Existing donation systems mainly focus on a single type of resource and do not provide intelligent matching.

¤ Therefore, there is a need for an AI-based platform that verifies users, predicts food demand/surplus, and intelligently matches donors with genuine receivers based on their needs and location.

OBJECTIVES...

Develop an AI-based platform to connect donors with genuine receivers.

Reduce food wastage by predicting food demand and surplus in hotels and restaurants.

Provide verified profiles for both donors and receivers to improve trust and reliability.

Match donations intelligently based on the receiver’s needs and location.

Support multiple donations such as food, clothing, electronics, financial aid, and educational assistance.

Send timely notifications to nearby verified receivers when resources are available.

LITERATURE REVIEW...

S.No.	Paper / Study	Year	Method / Technology	Major Findings	Limitation / Relevance to Proposed Work
1	**Machine-learning-based demand forecasting against food waste: Life cycle environmental impacts and benefits of a bakery case study**	2024	Machine Learning, Demand Forecasting	The study evaluated an ML-based forecasting service for reducing food waste in bakeries. The case study reported a substantial reduction in bakery returns through improved sales forecasting.	Focuses primarily on bakery demand forecasting and food-waste reduction. It does not provide an integrated donor–receiver platform for redistributing predicted surplus food.
2	**Smart Food-Sharing Platforms for Social Sustainability: A Heuristic Algorithm Approach**	2026	AI, Smart Contracts, Heuristic Algorithm	The study proposes smart food-sharing mechanisms and considers factors such as demand, donation rates, shelf life, transport costs, and expiration dates for redistribution decisions.	Concentrates mainly on food-sharing optimization. It does not combine food donation with clothing, electronics, financial assistance, and general beneficiary requirements.
3	**Why Do Hospitality and Food Service Businesses Use Food Sharing Platforms?**	2026	Digital Food-Sharing Platforms, Data Analytics	The study investigates food-sharing platforms as digital solutions for redistributing surplus food and identifies operational capabilities, inventory management, and data analytics as important factors for effective adoption.	Examines adoption and operation of food-sharing platforms but does not provide the proposed integrated AI-based donor–beneficiary matching ecosystem.
4	**From Waste to Worth: Enhancing Access to Edible Surplus Food Using a Mobile App for Better Distribution**	2025	Mobile Application, Geo-location, Real-Time Listings	The proposed application connects restaurants, supermarkets, households, and recipients using real-time food listings, pickup scheduling, and geo-location services.	Focuses specifically on edible surplus food and does not address clothing, electronics, monetary donations, or AI-based demand prediction.
5	**It Is a Match! How Donors and Nonprofit Organizations Come Together on a Matching Donation Platform**	2024	Online Donation Platform, Donor–Nonprofit Matching	The study examines matching donation platforms that align donors' preferences with nonprofit organizations and projects.	Concentrates on monetary/donation matching and does not address food surplus prediction, physical-item redistribution, or location-based urgent resource matching.

RESEARCH GAP...

Existing Approach	Existing Capability	Identified Gap	Proposed Project
Food-demand forecasting	Predicts future food demand	Does not directly connect predicted surplus with beneficiaries	Predicts demand and connects potential surplus with nearby verified receivers
Food-sharing platforms	Connect food donors and recipients	Mainly limited to food	Supports food, clothing, electronics, and monetary donations
Online donation platforms	Connect donors with charitable organizations	Limited visibility of local/remote requirements	Displays verified beneficiary requirements and recommends suitable donors
Location-based donation systems	Helps identify nearby donation opportunities	Matching may not consider multiple donation categories and urgency	Uses location, donation type, urgency, and requirements
Donor–nonprofit matching	Matches donors with organizations/projects	Mainly focused on donation preferences or financial contributions	Supports both monetary and physical-resource donations
Manual donation coordination	Enables donor and receiver communication	Time-consuming and may result in delayed distribution	Uses automated matching and notifications
Existing food redistribution systems	Redistribute available surplus food	Generally react after surplus is generated	Uses AI prediction to estimate demand and potential surplus before distribution

PROPOSED METHODOLOGY...

User Registration & Verification..

¤ Separate registration for donors and beneficiaries.

¤ Includes NGOs, schools, orphanages, old-age homes, and businesses.

¤ Basic details and contact verification.

Donor Module..

- List available resources:
- Food
- Clothing
- Electronics
- Educational materials
- Household items
- Monetary assistance
- Provide quantity, condition, location, and availability.

Receiver Module

Verified receivers can post requirements.

Specify required resource and quantity.



AI Prediction & Intelligent Matching...

AI Food Demand Prediction..

- Uses historical restaurant data.
- Predicts future food demand using ML.
- Estimates possible surplus food.

Flow : Data → Preprocessing → ML Model → Demand → Surplus.

Donor–Receiver Matching..

- Matches suitable donors and receivers.
- Considers: ¤Location
  ¤Category
  ¤Urgency
  ¤Quantity
  ¤Availability

NOTIFICATIONS, LOCATION & TRACKING..

Notifications.

¤Automatic alerts to donors and receivers.

¤Reduces manual searching.

Location Matching.

¤Finds nearby suitable matches.

¤Prioritizes urgent food donations.

Donation Tracking.

¤Records donation details and status.

¤Tracks the complete process.

Flow:

Posted → Matched → Accepted → Collected → Delivered → Completed

PROPOSED SYSTEM ARCHITECTURE

1 User Registration & Verification

- Separate registration for donors, NGOs, schools, orphanages, old-age homes, businesses, and beneficiaries
- Basic details and contact verification ensure authentic profiles

2 Donor Module

- Donors can list: Surplus food, Clothing, Electronics, Educational materials, Household items, Monetary assistance
- Provide quantity, condition, location, and availability

3 Receiver Requirement Module

- Verified receivers can post their requirements
- Examples: meal packets, clothing, laptops, financial assistance
- Requirements are stored for intelligent matching

4 AI-Based Food Demand Prediction

- Uses historical restaurant data (customers, food prepared/sold, day, season, festivals, events, etc.)
- ML model predicts expected food demand and estimates potential surplus

Historical Data → Preprocessing → Feature Selection → ML Model → Demand Prediction → Surplus Estimation

5 Intelligent Donor–Receiver Matching

- Matches donors and receivers based on:

Location | Category | Urgency | Quantity | Availability | Receiver Priority

- Recommends the most suitable donors and receivers

6 Notification Module

- Sends automatic alerts to matched donors and receivers
- Reduces the need for continuous manual searching

Donor Notification ← → Receiver Notification

7 Location-Based Matching

- Uses donor and receiver locations to identify nearby matches
- Gives priority to nearby receivers for time-sensitive resources such as food

Donor — Receiver

8 Donation Tracking

- Records donor, receiver, resource type, quantity, date and time, status, pickup/delivery information
- Ensures transparency and allows donors to view history

Posted → Matched → Accepted → Collected → Delivered → Completed

SMART DONATION

Software / Technology Stack
(Difference)

Software / Technology	Purpose
HTML	Web page structure
CSS	User interface styling
JavaScript	Frontend interaction
Python	Backend and AI/ML development
Flask / Django	Backend framework
Pandas	Dataset processing
NumPy	Numerical operations
Scikit-learn	Machine-learning model development
MySQL / Firebase	Database management
Google Maps API / OpenStreetMap	Location and distance services
VS Code	Development environment
Git / GitHub	Version control
Android Studio *(if required)*	Mobile application development

DATASET..

The dataset for Smart Donation Connect contains information related to food demand, donations, and beneficiary requirements. It includes details such as date, day, number of customers, food prepared, food sold, surplus food, donation type, quantity, donor/receiver location, and urgency level.

Historical restaurant data is used to train the Machine Learning model for predicting food demand and identifying possible surplus food. The dataset also helps the system match available donations with suitable receivers based on their requirements, location, and urgency.

IMPLEMENTATION..

The system is divided into the following modules:

1. User Registration & Login Module

Allows donors and receivers to create accounts and securely access the system.

2. Donor Module

Allows users to add food, clothes, electronic items, or monetary donations.

3. Receiver Module

Allows verified beneficiaries to post their requirements and required quantity.

4. Verification Module

Helps verify donor and receiver details to improve trust and reliability.

5. AI Food Prediction Module

Uses historical food-demand data to predict the expected number of customers and possible surplus food.

6. Smart Matching Module

Matches available donations with suitable beneficiaries based on donation type, location, quantity, and urgency.

7. Notification Module

Sends alerts to donors and receivers when a suitable donation or requirement is available.

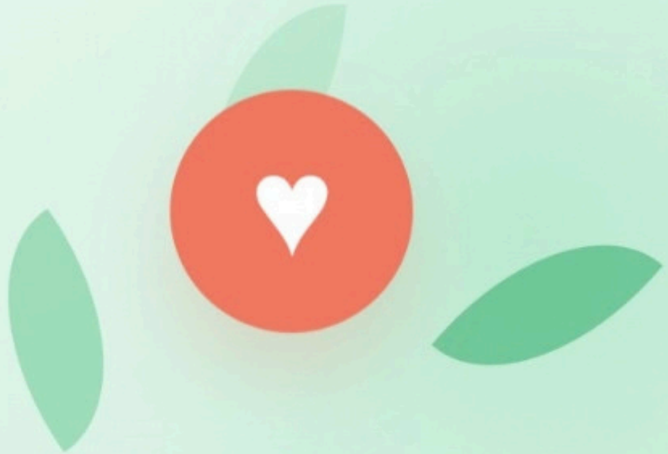
8. Donation Tracking Module

Tracks the donation from posting to completion.

Dashboard - Smart Donation C | (38) WhatsApp | Login - Smart Donation Connec

127.0.0.1:5000/login

Smart Donation
Connect



Together we can build a better tomorrow.

Every donation creates a connection. Every connection creates an impact.

WELCOME BACK

Welcome Back

Login to your account

Email Address

Enter your email

Password

Forgot password?

Enter your password

Remember me

Login

Don't have an account? Register

← Back to Home

(22) YouTube | Donations | kisskh.pl | Post Donation | Add shortcut

Type here to search

35°C

ENG
US

14:51
28-09-2026

Dashboard - Smart Donation C | (38) WhatsApp

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Smart Donation Connect

Profile | Notifications | Logout

Dashboard

- Dashboard
- Donate
- Post Donation
- Requirements
- Matches
- Notifications
- Profile
- Settings

Welcome Kirthana K

Thank you for being part of Smart Donation Connect. Together, we can connect donations with people who need them.

Donate

Share food, clothes, electronics, books, money or other useful items.

Post Donation

Requirements

View current requirements from verified beneficiaries.

View Needs

Matches

View donation and beneficiary matches.

View Matches

Recent Donations

View All

No donations have been posted yet.

(22) YouTube | Donations | kisskh.pl | Post Donation | Add shortcut

Type here to search

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28-09-2026

FUTURE ENHANCEMENTS..

The system can be enhanced in the future with:

- Mobile application for Android and iOS.
- More advanced AI models for accurate food-demand prediction.
- Real-time GPS-based donor–receiver matching.
- Online payment integration for monetary donations.
- QR-code-based donation verification.
- Automated pickup and delivery coordination.
- Multilingual support for different Indian languages.
- Integration with NGOs and government welfare organizations.
- AI-based fraud and fake-profile detection.
- Real-time analytics dashboard.
- reputation and contribution history.
- Cloud deployment for nationwide scalability.

CONCLUSION..

The Smart Donation Connect project provides an AI-enabled approach to connecting donors with people and organizations in need. The system combines donation management, beneficiary requirements, location-based matching, verification, notifications, and AI-based food-demand prediction in a single platform.

The proposed solution can help reduce food wastage, encourage the reuse of clothes and electronic items, and improve the visibility of genuine needs in underserved communities. By intelligently matching available resources with verified requirements, the system aims to make the donation process more efficient, transparent, and accessible.

Thank You